

Jorge Alfredo Robles Calderón

Physicist — experimental materials, superconductivity and optics

Trinidad, Casanare / Bogotá, Colombia · open to remote, hybrid or relocation

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SUMMARY

Experimental physicist with independent experience in scientific software. Synthesis and characterization of spinel ferrites; a three-dimensional two-band superconductor solver designed, written and validated from scratch; early work in first-principles nonlinear optics and low-cost computational optics; web and data applications deployed and in use. B.Sc. Physics, Universidad Nacional de Colombia, degree conferred 5 November 2026.

Areas: condensed matter physics — superconductivity, functional materials, optics. Open to funded master's or PhD positions, and to research or industry roles in scientific software, data and materials R&D.

RESEARCH

Three-dimensional two-band TDGL solver — independent research, UNAL, 2025–present

Design, implementation and validation of a 3D two-band time-dependent Ginzburg–Landau solver for mesoscopic superconductors under field and transport current: gauge-invariant link variables, self-consistent vector potential, self-field current injection, Josephson voltmeter. Octave reference v13.3; Julia solver bit for bit against it; about 56 000 test assertions; lowest Landau level within 1.6 % and H_{c3}/H_{c2} within 0.05 %. Released as open source: **juliaTDGL 1.0** (MIT), github.com/georgeotjg/juliaTDGL.

Silent failure modes in three-dimensional two-band time-dependent Ginzburg–Landau simulations of geometric superconducting diodes — J. A. Robles Calderón, preprint, Zenodo (2026), [doi:10.5281/zenodo.23163207](https://doi.org/10.5281/zenodo.23163207). Eighteen failure modes, each yielding a converged but wrong quantity, each with a test; symmetry-exact diode control; diode efficiency bracketed from –47 % to –21 % [–33, –7] across padding, no converged value claimed. Currently: the diode at MgB_2 -like parameters, with Joule heating and interband drag.

$\text{Zn}_{1-x}\text{Mg}_x\text{Fe}_2\text{O}_4$ spinel ferrites — bachelor's thesis (trabajo de grado), UNAL, 2026 (152 pp)

Advisor: D. A. Landínez Téllez. Synthesis of the series by glycine-assisted combustion; XRD with Rietveld refinement, SEM/EDX, UV-Vis diffuse reflectance. Analysis automated in Python, including band-gap fits whose window is selected by robustness (scikit-learn). Single-phase $Fd\bar{3}m$ solid solution, coherent domains of 20–26 nm, direct optical gap 1.88–1.91 eV nearly independent of composition. GGA+U band structures (treatment proposed and analysed in the thesis, computed within the research group) place the gap in the Fe–O subsystem.

Nonlinear optics of Weyl semimetals — started 2026

First-principles photocurrents in Weyl semimetals: Octopus (real-time TDDFT) and Quantum ESPRESSO + Wannier90 + Julia. Real-time lattice model reproducing the quantized circular photogalvanic response; Octopus validated on silicon including a symmetry null test; GaAs Wannier chain running.

Computational lensmeter — design stage

Measurement of spectacle lenses (sphere, cylinder, axis) by transmission deflectometry with a phone camera's RAW sensor and a laptop screen; elliptic inverse solver and thick-lens model.

APPLIED PROJECTS

Geospatial field-data platform. GPS-tagged forms, maps and spatial queries; Next.js, Supabase (PostgreSQL/PostGIS), Leaflet, PWA with offline queue. Deployed and used in the field for interviews, surveys and territorial records.

OCR + local language model. National Astronomical Observatory, UNAL: metadata from scanned documents, with validation that rejects malformed extractions.

Shape editor and run manager for the superconductor solver (JavaScript, Python): 3D sample shapes with live symmetry checks; terminal menu to launch, follow and resume long runs.

Commercial catalogue site. Next.js, TypeScript and Tailwind on Vercel, quotations over WhatsApp. tecnimotos-villamil.vercel.app

EXPERIENCE

Administrative and Technical Assistant — National Astronomical Observatory, Universidad Nacional de Colombia · 2023–2026

Extension and Technology-Transfer Assistant — Department of Physics, Universidad Nacional de Colombia · 2024–2025

Academic and Scientific Support — Universidad Nacional de Colombia · 2022–2025

Customer Operations — Iké Asistencia and prior roles · 2018–2023

EDUCATION

B.Sc. Physics — Universidad Nacional de Colombia, Bogotá · 2019–2026. Degree conferred 5 November 2026. Bachelor's thesis graded 5.0/5.0; GPA 4.0/5.0.

Technical degree, Administrative Assistance — SENA · 2019

SKILLS

Computation: Python (NumPy, SciPy, pandas, scikit-learn, PyTorch, Matplotlib), Julia, MATLAB/Octave, SQL, TypeScript, LaTeX, Git, Linux.

Numerical methods: finite differences on staggered grids, gauge-invariant discretisation, IMEX schemes, Poisson/elliptic solves, preconditioned Krylov solvers, regression and acceptance testing.

First principles: DFT/DFT+U concepts and band-structure analysis; Quantum ESPRESSO and Octopus (real-time TDDFT), in current use.

Laboratory: combustion synthesis, XRD with Rietveld refinement, SEM/EDX, UV-Vis diffuse reflectance.

Data and cloud: data analysis and fitting pipelines in Python; SQL (PostgreSQL), Apache Spark, AWS, Metabase dashboards; containers (Docker/Podman).

GIS: PostGIS, spatial queries, Leaflet, GPS field data.

Web: Next.js/TypeScript, Supabase (deployed applications).

Languages: Spanish (native), English (B2).